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Bachelor Thesis

Design and Characterization of an Embedded Micropump-Driven Generator for Gel-Droplets

under the supervision of

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Abstract

This thesis presents the design and characterization of an embedded micropump-driven microfluidic system capable of generating aqueous and gel-based droplets. A modular platform was developed that integrates custom control electronics for piezoelectric micropumps, a compact housing, and a hybrid droplet generator composed of a 3D-printed inlet block and laser-patterned adhesive-tape microchannels. The system was designed to enable low-flow operation, portability, and compatibility with a variety of fluidic conditions.

Comprehensive characterization of the micropumps was performed using distilled water, providing reproducible measurements of pressure and flow as a function of driving voltage and frequency. Additional qualitative tests with higher-viscosity media—rapeseed oil and carboxymethyl cellulose (CMC) gels—were used to evaluate operational limits and pumping capability under increased hydraulic load. Droplet formation experiments employed rapeseed oil as the continuous phase and either water or CMC solutions up to 1 wt% as the dispersed phase. Stable droplet generation was successfully demonstrated for the two-inlet design, while the three-inlet configuration revealed challenges related to asynchronous pump behavior and mismatched hydraulic resistances.

The results confirm that piezoelectric micropumps can reliably drive a microfluidic droplet generator and enable controlled production of water and gel droplets. However, performance was strongly influenced by pump-to-pump variability, residual air inclusions, and transient pressure peaks.

Finally, the thesis outlines several avenues for further development, including closed-loop flow and pressure control, in-situ electrical characterization for resonance-based optimization, and the implementation of a fully standalone pump-driver module to enhance usability and long-term stability.

Contents

Abstract **1**

1 Introduction **3**

2 Concept **5**
2.1 Micropump and Control Electronics 5
2.1.1 Micropump Specifications 5
2.1.2 Lowdriver Circuit and Functionality 6
2.1.3 Custom PCB Design 6
2.1.4 Control Software and Web Interface 8
2.2 Housing and Mechanical Design 9
2.3 Droplet Generator Fabrication 10

3 Methods **12**
3.1 Power Consumption Measurement 12
3.2 Micropump Characterization 12
3.2.1 Pressure Measurement 12
3.2.2 Flow Measurement 13
3.3 CMC Gel 13
3.4 Droplet Generation and Experiments 14

4 Results **16**
4.1 Power Consumption Measurement 16
4.2 Pressure Measurement 17
4.3 Flow Measurement 18
4.4 Droplet Generation 19

5 Discussion and Conclusion **21**
5.1 System Behavior and Practical Considerations 21
5.2 Summary of Findings 22
5.3 Outlook 22

A Appendix **25**
A.1 System Overview 25
A.2 Validity of Flow Measurement 26
A.3 Mainboard for stand-alone pumps 27
A.4 Driver schematic 28

Introduction

Microfluidics has emerged as a key enabling technology for the manipulation of small fluid volumes, with applications ranging from biomedical diagnostics to advanced materials processing. Among its approaches is droplet-based microfluidics, where discrete and monodisperse droplets serve as controllable reaction vessels. These systems offer significant advantages, including precise dosing, high throughput, and reduced reagent consumption [3, 10].

A central challenge in droplet microfluidics is the reliable and tunable generation of droplets. Various passive and active methods have been developed, including flow-focusing geometries, T-junctions, and piezo-actuated droplet-on-demand systems [8, 3]. Active approaches are particularly attractive when dynamic control over the droplet size, frequency, or timing is required. In this context, micropumps based on piezoelectric actuation have received considerable attention due to their compact form factor, fast response, and compatibility with miniaturized systems [10].

In parallel, the use of hydrogel-forming materials such as carboxymethyl cellulose (CMC) has enabled new possibilities for generating and studying soft, biocompatible droplets. CMC-based gels can form stable droplet structures without the need for photopolymerization, making them particularly suitable for applications that require mild processing conditions or biological compatibility. Such gel droplets can be used as model systems for encapsulation, controlled release, or material transport in microfluidic environments.[7]

The present work addresses the design and characterization of an embedded micropump -driven droplet generator dedicated to the formation of CMC-based gel droplets. The system integrates custom-built electronic control for piezoelectric micropumps, a compact housing, and a 3D-printed droplet generator. Experimental characterization includes evaluation of pumping performance, droplet size distribution, and the influence of operating parameters on droplet stability and formation behavior.

By combining a cost-effective and modular design with precise droplet control, this study contributes to the ongoing efforts to make droplet-based microfluidic technologies more accessible and adaptable to a wide range of applications.

Building on this motivation, the present work is guided by three central research questions:

1. Can a micropump be used to generate and control microfluidic gel droplets?
2. How can such a micropump–droplet generator system be designed and built?
3. What are the limits of this system in terms of droplet size, stability, and overall performance?

Addressing these questions provides both a theoretical foundation and an experimental validation for the design and characterization of the embedded micropump-driven generator developed in this thesis.

Concept

2.1 Micropump and Control Electronics

The fluidic actuation in this work relied on commercial piezoelectric diaphragm micropumps (mp6, Bartels Mikrotechnik GmbH, Germany). Each micropump was operated through a dedicated lowdriver integrated circuit supplied by the manufacturer. To enable flexible and scalable operation, a custom printed circuit board (PCB) was designed using KiCAD and fabricated by JLCPCB. This PCB integrates an ESP32 wireless microcontroller (Espressif Systems), which also serves as a host for a web-based interface. Through this interface, pump parameters such as actuation frequency and voltage amplitude can be configured wirelessly. Both the firmware and the web application were specifically developed for this project.

2.1.1 Micropump Specifications

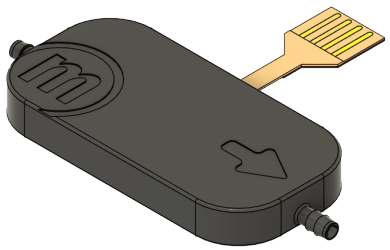


Figure 2.1: Bartels - mp6 piezo micropump

The micropumps employed in this work were commercial piezoelectric diaphragm pumps (mp6, Bartels Mikrotechnik GmbH, Germany). Owing to their compact size ($30 \times 15 \times 3.8$ mm^3) and low weight (2 g), the pumps can be easily integrated into portable microfluidic systems. Each unit features two piezoelectric actuators and consumes less than 200 mW of electrical power, making them suitable for battery-powered applications. The fluidic connectors are barbed tube clips with an outer diameter of 1.9 mm, while electrical interfacing is achieved through a 1.25 mm pitch flex connector.

The pumps are self-priming and compatible with liquids, gases, and liquid–gas mixtures within an operating temperature range of 0–70°C. The typical lifetime is specified as 5000 h under standard conditions.

In terms of hydraulic performance, the mp6 provides a maximum water flow rate of approximately 7 mL/min with a back pressure tolerance up to 600 mbar. For gaseous media, the maximum flow rate is around 18 mL/min with a back pressure limit of 100 mbar. The pump body is made of polyphenylene sulfone (PPSU), which is chemically stable against many aqueous solutions but not resistant to concentrated alcohols such as methanol or ethanol. Overall, the mp6 micropumps offer a reliable and energy-efficient solution for microfluidic droplet generation and other low-flow actuation tasks.

2.1.2 Lowdriver Circuit and Functionality

The lowdriver is a dedicated integrated circuit designed to operate piezoelectric micropumps such as the mp6 series. It provides a straightforward interface to the piezo actuators and communicates via the I^2C protocol using the SDA and SCL pins, allowing precise control of pump actuation.

The mp-Lowdriver controller (Bartels Mikrotechnik GmbH, Germany) features compact dimensions of 11.9 × 27.3 × 6.3 mm and is capable of adjusting key operational parameters including amplitude, frequency, and signal shape. The output amplitude can be set between 0 and 150 V_{pp} in 255 discrete steps, while the frequency can be selected from 8 to 2000 Hz in increments of 7.8125 Hz. The signal can be generated as a sine wave or a user-defined custom waveform. The device operates from a 3–5.5 V DC supply, with an absolute maximum rating of 6 V. The pin configuration follows a DIL-18 format with a horizontal pitch of 2.54 mm and vertical spacing of 7.62 mm.

Although the mp-Lowdriver can technically reach frequencies up to 2000 Hz, the mp6 micropump has been validated only up to 800 Hz. Frequencies above this threshold are not recommended to ensure reliable operation and to avoid potential damage to the pump.

2.1.3 Custom PCB Design

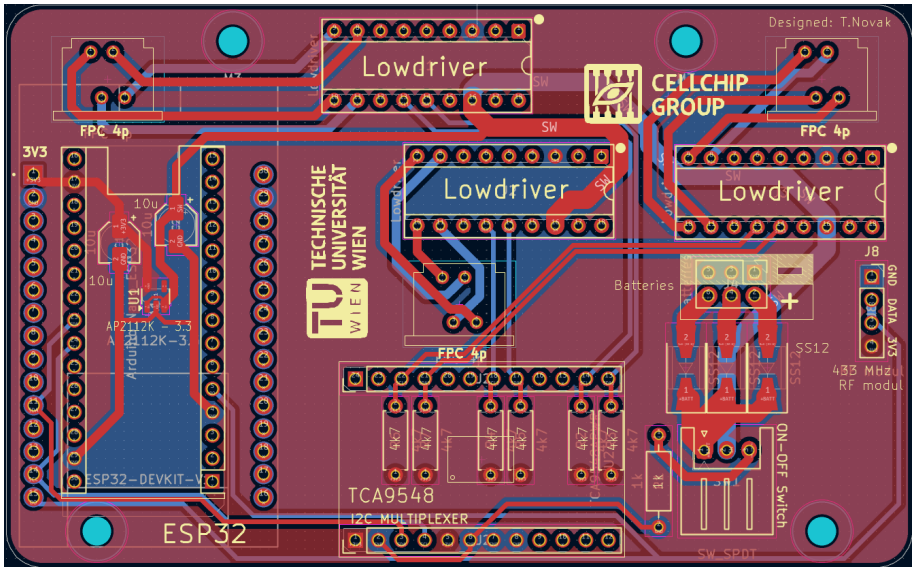


Figure 2.2: PCB footprint

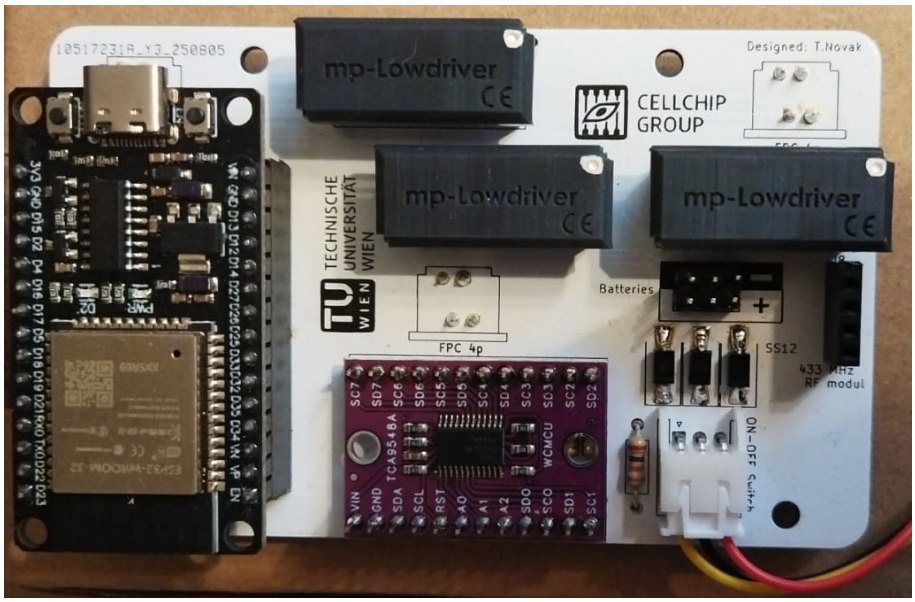


Figure 2.3: Assembled PCB

Power Supply and Regulation

The custom PCB is equipped with connectors for up to three rechargeable lithium-ion batteries, each rated at 10 000 mAh and 3.7 V nominal voltage. Although the design allows the use of up to three cells, in the current prototype only a single battery is installed and used to power the system.

To prevent equalization currents between the batteries when multiple cells are connected, each battery input is protected by an individual Schottky diode. A two-position mechanical switch is placed after the diodes to physically disconnect the power source from the entire circuit, ensuring that no quiescent current is drawn in the off state.

For battery charging, a TP4056 charging module with a USB-C connector is integrated into the system. It supports a maximum charging current of 1 A and includes built-in thermal regulation, overcharge protection, and additional safety circuitry to protect both the battery and the electronics during operation.

The regulated supply voltage for the digital and analog circuitry is generated by an AP2112K linear regulator, which provides a stable 3.3 V output from the battery input. Stability of the regulator is further improved by two decoupling capacitors placed at the input and output, reducing supply ripple and ensuring reliable operation of the microcontroller and driver circuits.

Additional filtering capacitors are distributed across the PCB to minimize voltage transients and high-frequency noise generated by the switching components of the micropump driver. This power architecture ensures stable operation under dynamic load conditions while maintaining safety and modularity for future system extensions.

Microcontroller and Communication

The control of the micropumps is managed by a dedicated microcontroller. For flexibility, the PCB was designed to accommodate two interchangeable options: either an Espressif ESP32-WROOM-32U (30-pin version) or an Arduino Nano ESP32 module. Both microcontrollers provide integrated Wi-Fi connectivity, enabling wireless communication and remote configuration of operating parameters.

In addition to the microcontroller, a 433 MHz RF transceiver module was also included in the design. This low-power communication interface allows future extensions of the system for remote control applications where Wi-Fi connectivity may not be available or where reduced power consumption is required.

Driver Integration

To maintain modularity and simplify replacement, the Bartels lowdriver ICs were not soldered directly onto the PCB but instead mounted in DIL-18 sockets. The driver modules communicate with the microcontroller via the I^2C protocol. To support multiple devices on the same bus, a TCA9848 I^2C multiplexer was implemented, with 4.7 kΩ pull-up resistors ensuring reliable signal integrity.

The connection between the drivers and the micropumps is realized using 4-pin flat-flex (FPC) connectors, providing a compact and robust interface for signal and power delivery.

2.1.4 Control Software and Web Interface

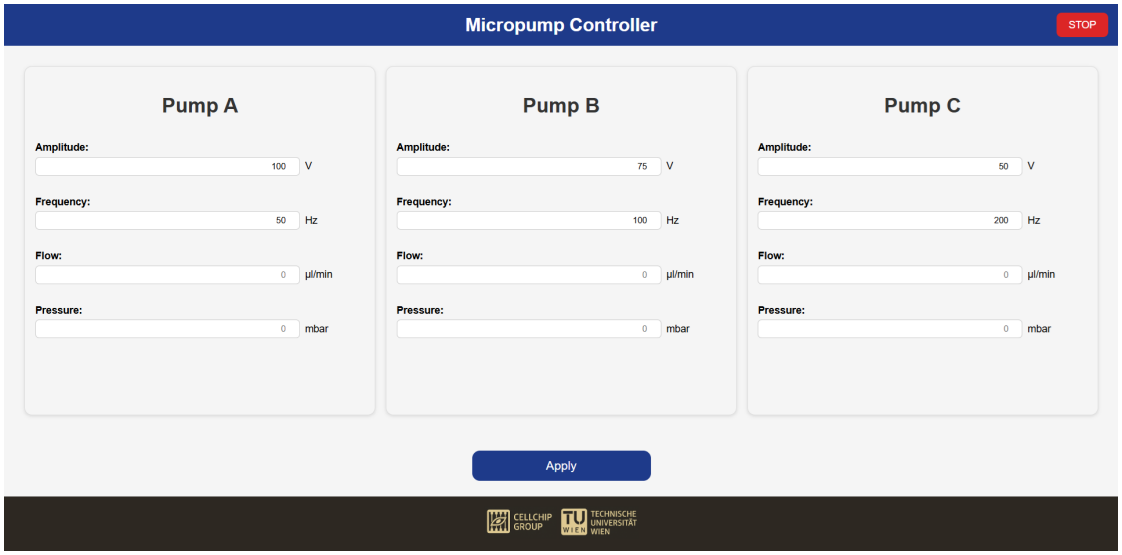


Figure 2.4: Web interface

To enable contactless configuration, a web-based control interface (Fig. 2.4) was implemented and hosted directly on the ESP32 microcontroller. Once connected to the Wi-Fi network established by the device, users can access the web application through a standard browser. The interface follows a minimalist design and provides independent control of the three micropumps, allowing adjustment of actuation voltage and frequency, as well as the option to stop all pumps when required.

In addition to the current functionality, the software architecture offers the possibility of future extensions, such as the integration of flow rate and pressure estimation, further enhancing the usability of the system in microfluidic applications.

2.2 Housing and Mechanical Design

Mechanical Design and Housing

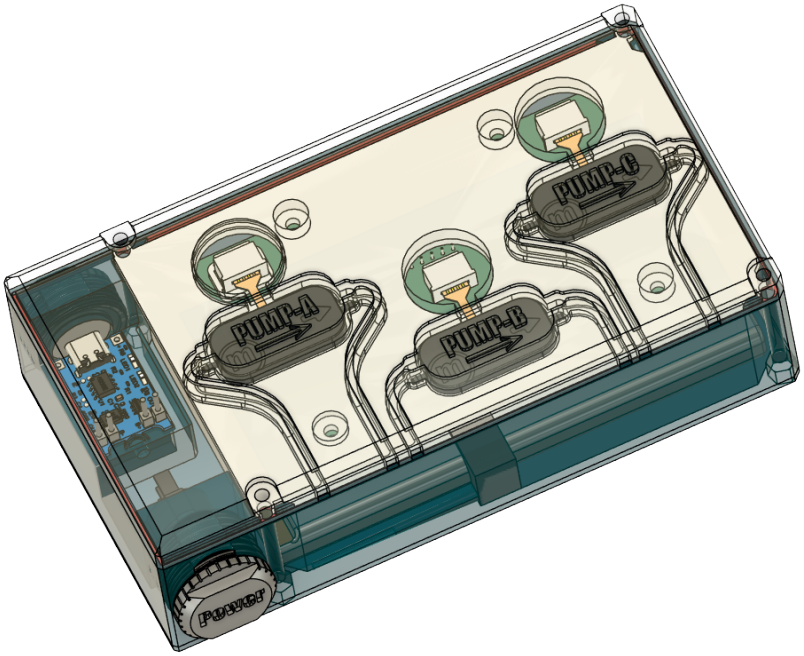


Figure 2.5: Housing

To enable safe and compact operation, a dedicated housing (Fig. 2.5) was designed in Autodesk Fusion 360. The enclosure was optimized for straightforward integration into a laboratory incubator and was fabricated using PETG filament with a 0.25 mm nozzle diameter to ensure maximum gas tightness. All structural walls have a minimum thickness of 2 mm, providing sufficient rigidity and impact resistance. The complete device has an external dimensions of 154 mm in length, 95 mm in width, and 42 mm in height. The resulting device offers mechanical protection, modularity, and ease of handling.

The housing consists of three structural levels. The bottom compartment accommodates the rechargeable battery, the TP4056 charging module with an external USB-C charging port, and the main power switch for complete system isolation. Two threaded sealing caps are integrated to provide moisture protection for the USB charging port and the power switch when the device is used in incubator. An optional custom 1 mm thick TPU-based sealing ring can be inserted between the lower housing and the top cover to provide additional gas-tight insulation around the enclosure. A retaining strap for the battery is also included to prevent displacement during handling or transport.

The middle level holds the three micropumps with their tubing connections and provides a mounting interface for the PCB. This arrangement minimizes the required wiring and simplifies assembly. Finally, the top cover serves as both a protective shield and an insulating barrier, mechanically safeguarding the pumps and reducing environmental exposure. The cover also includes printed labels to clearly indicate the position of each pump.

All three sections are secured together using four long M3 through-bolts (40 mm length) that pass through the entire structure, ensuring a rigid and durable assembly

suitable for repeated laboratory use.

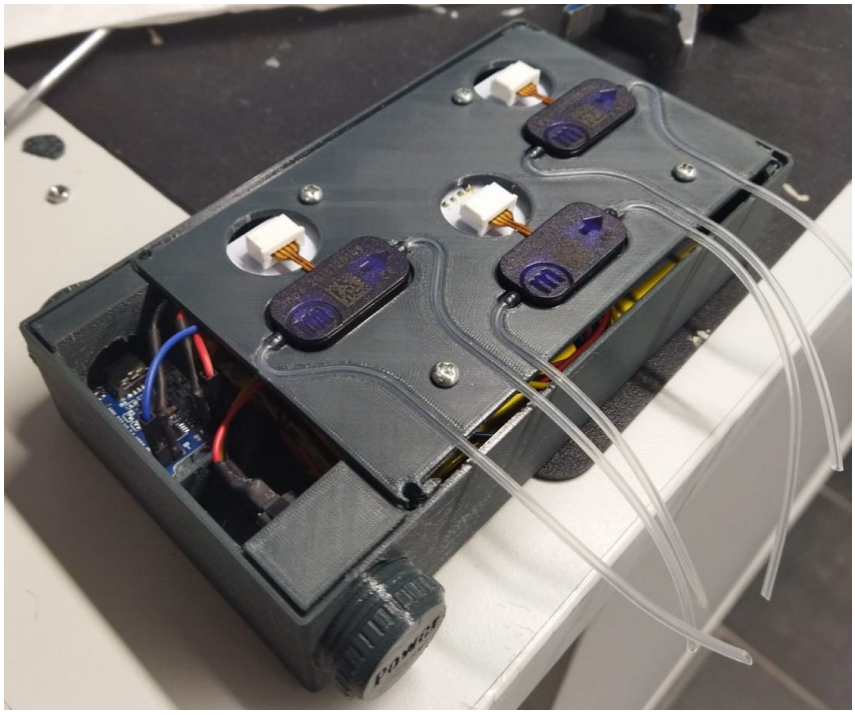


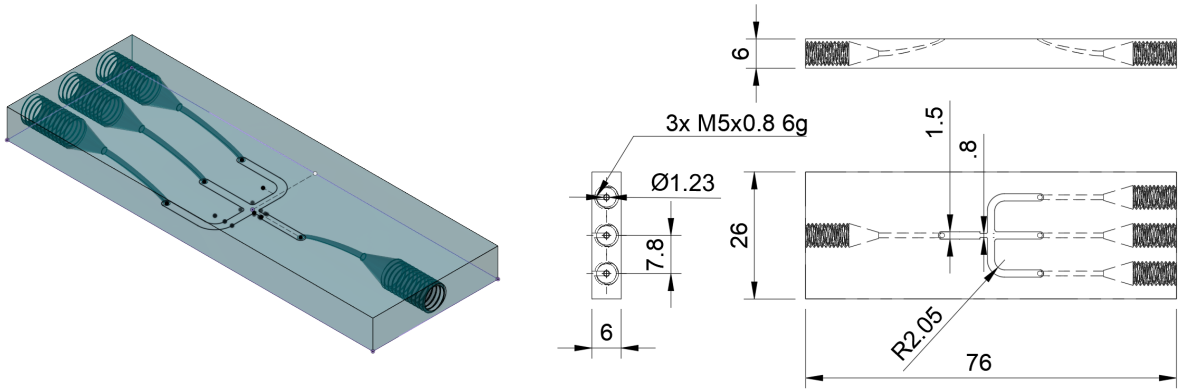
Figure 2.6: Assembled Housing

2.3 Droplet Generator Fabrication

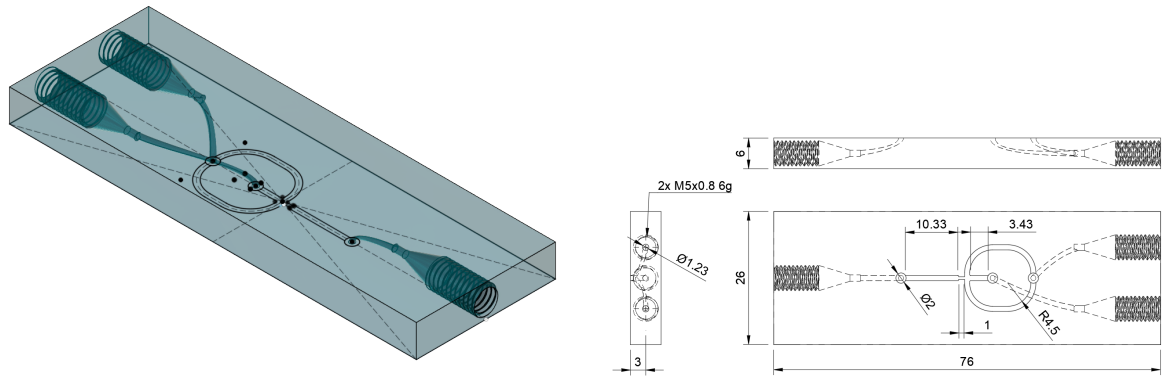
A standalone microfluidic droplet generator (Fig. 2.7) was developed to interface directly with the micropump module. Two versions were produced—one with two inlets and one with three inlets—both following the same general fabrication workflow. In contrast to fully 3D-printed microfluidic devices, the present design separates the structural alignment features from the functional microchannels.

The inlet block, which positions the tubing connectors relative to the flow junction, was fabricated using high-resolution resin 3D printing. This printed component ensures precise alignment between the external Fluigent F-type connectors and the downstream microchannel layout. Depending on the experimental requirements, either Asiga PlasClear resin (optimized for fluorescence imaging but not biocompatible) or Formlabs BioMed Clear resin (biocompatible but less suitable for fluorescence microscopy) was used for printing.

The actual microchannels responsible for droplet formation were not printed. Instead, they were defined using double-sided adhesive tape (ARCARE 90106NB, Adhesives Research Inc., 150 µm thickness), which was laser-cut to the desired geometry using a Trotec Speedy 100 laser cutter. The adhesive layer was patterned to form the complete channel network, including the junction region. After cutting, the tape was laminated between a standard microscope glass slide on one side and the 3D-printed inlet block on the other, creating fully enclosed, optically accessible microchannels.



(a) 3D model (b) Sketch



(c) 3D model (d) Sketch

Figure 2.7: Overview of the two droplet generators: 3-inlet and 2-inlet designs.

Methods

To evaluate the functionality and long-term applicability of the developed micropump system, a series of characterization experiments were conducted. The measurements aimed to assess electrical consumption, pumping performance under varying operating conditions, as well as the capability of the system to generate stable microdroplets in a microfluidic setup.

3.1 Power Consumption Measurement

In order to determine the suitability of the system for long-term and modular use, the electrical power consumption was measured using a digital multimeter. The current was recorded by inserting the meter directly between the battery and the anode of the first Schottky diode. From these data, the average power draw was calculated, providing an estimate of operational lifetime and integrability into portable setups.

3.2 Micropump Characterization

The performance of the Bartels MP6 piezoelectric diaphragm micropumps was evaluated with respect to two primary control parameters: the applied peak-to-peak voltage (V_{pp}) and the driving frequency (f). These micropumps operate based on a piezoelectric diaphragm that cyclically deforms under an applied voltage, generating a pressure differential that drives fluid through the microchannels. Integrated passive valves ensure unidirectional flow, allowing precise and repeatable fluid pumping.

All characterization measurements were carried out using distilled water to ensure accurate, stable, and reproducible pressure and flow data. Higher-viscosity fluids, such as rapeseed oil and CMC gels, were tested only qualitatively to verify that the pumps remained operational under increased hydraulic load. These tests served solely to assess pumpability; no quantitative characterization was performed with these fluids based on the constrains of the flow measurement devices.

3.2.1 Pressure Measurement

Two key measurements were conducted: the relationship between generated pressure and applied voltage (P vs. V_{pp}), and the temporal pressure response at different driving frequencies (P vs. t). For all experiments, a 110 mm PTFE tube was installed between the pump inlet and the pressure sensor to standardize the hydraulic resistance.

Pressure values at different operating conditions were measured using a Fluigent Pressure M sensor, and all data were recorded with the Fluigent OXYGEN software. This setup provided direct insight into the achievable pressure ranges and enabled optimization of the pumping parameters across different voltages, frequencies, and fluid viscosities.

3.2.2 Flow Measurement

Similar to the pressure characterization, flow rate measurements were performed to evaluate the pump performance under various operating conditions. The experiments focused on determining the relationship between flow rate and applied peak-to-peak voltage (Q vs. V_{pp}), as well as the temporal flow behavior at different driving frequencies (Q vs. t).

A Fluigent Flow Unit M sensor (maximum range: 80 µL/min) was used to measure the flow rate with high temporal resolution. All data were acquired and recorded using the Fluigent OXYGEN software. The same experimental setup was maintained as for the pressure tests, including the 110 mm PTFE tube between the pump outlet and the flow sensor, ensuring consistent hydraulic conditions.

These measurements provided complementary information to the pressure data, allowing a comprehensive assessment of the micropump's dynamic response and efficiency at different voltages, frequencies, and fluid viscosities.

3.3 CMC Gel

To evaluate the influence of fluid viscosity on micropump performance, aqueous solutions of carboxymethyl cellulose (CMC) were prepared at concentrations of 0.2, 0.5, 1.0, 1.5, and 2.0 wt%. CMC is a water-soluble polysaccharide derivative commonly used as a rheological modifier due to its high viscosity and pronounced shear-thinning behavior. Shear-thinning refers to the decrease in apparent viscosity with increasing shear rate, meaning that the fluid flows more readily under higher mechanical stress. This non-Newtonian characteristic is particularly relevant in microfluidic applications, where flow conditions can vary rapidly within confined geometries.

For the characterization of the micropump, the zero-shear viscosity (μ_0) was chosen as the reference parameter. The reason is that μ_0 represents the viscosity of the fluid under minimal shear, corresponding to the conditions in the microchannels before significant flow occurs. It provides a baseline for comparing fluids of different concentrations and allows estimating the pump's capability to initiate flow in highly viscous solutions [1].

Table 3.1: Zero-shear viscosities (μ_0) of CMC solutions from Benchabane and Bekkour [1].

CMC concentration (wt%)	μ_0 (Pa · s)
0.2	0.019
0.4	0.045
1.0	0.530
1.5	1.970
2.0	6.355

Each solution was allowed to rest for 24 hours after preparation to ensure complete dissolution and homogenization. This equilibration period was necessary to achieve stable

viscosity characteristics across all concentrations. The progressive increase in μ_0 with concentration demonstrates why higher-concentration CMC solutions demand higher pressure differentials or optimized pumping parameters to maintain consistent flow in microfluidic channels.

3.4 Droplet Generation and Experiments

The droplet generation process was investigated under the previously characterized pumping conditions. The primary objectives of this study were to analyze the physical mechanisms governing droplet formation and to identify the key parameters influencing droplet size and generation frequency. Optical analysis of the generated droplets was conducted using an BRESSER MikroCam PRO HDMI microscope, enabling precise determination of droplet diameters and formation rates.

According to the literature, droplet-based microfluidic systems rely fundamentally on the balance of interfacial forces acting between a dispersed and a continuous phase confined within microscale geometries [2]. Christopher and Anna [2] describe three principal configurations for droplet generation—co-flow, cross-flow, and flow-focusing—and emphasize that the breakup mechanism is primarily governed by the capillary number, viscosity ratio, interfacial tension, and geometric confinement.

In the present configuration, the formation of aqueous or gel-based droplets within a continuous oil phase can be classified within the *squeezing* or *dripping* regime. In this regime, the dispersed phase progressively penetrates into the junction until the pressure of the continuous phase builds up sufficiently to induce pinch-off. The relevant dimensionless parameters that describe this behavior are defined as:

$$\mathrm{Ca} = \frac{\mu_c U_c}{\sigma}, \qquad \lambda = \frac{\mu_d}{\mu_c},$$

where μ_c and μ_d denote the dynamic viscosities of the continuous and dispersed phases, U_c is the characteristic flow velocity of the continuous phase, and σ represents the interfacial tension. The capillary number (Ca) thus quantifies the relative contribution of viscous stresses to surface tension, while the viscosity ratio (λ) describes the contrast between the two fluids.

In this work, rapeseed oil served as the continuous phase. Typical viscosities of vegetable oils lie in the tens to hundreds of mPa · s range, and interfacial tensions against water are reported around 22–26 mN/m [5]. These properties, combined with the low Reynolds number of the flow, ensure operation in the laminar and capillary-dominated regime. The dispersed phase, either water or a 1 wt% CMC (carboxymethyl cellulose) solution, exhibited significantly different rheological properties. While water has a viscosity of approximately 1 mPa · s, the CMC solution may reach several hundred mPa · s at low shear rates (Tab. 3.1), leading to a substantially higher viscosity ratio λ.

The characteristic detachment time of a droplet, known as the pinch-off time (t_p), is determined by the balance of viscous, capillary, and inertial forces at the interface [2]. Depending on the dominant force balance, two primary regimes can be distinguished.

In the *viscous-dominated* regime ($\mathrm{Ca} < 10^{-2}$), typical of microfluidic systems operating at low Reynolds numbers, the breakup process is governed by the competition between viscous stresses and surface tension. The characteristic time can be estimated as:

$$t_p \sim \frac{\mu_d r_0}{\sigma},$$

where r_0 is the characteristic radius of the channel. Accordingly, an increase in dispersed-phase viscosity or a decrease in interfacial tension prolongs t_p, leading to slower detachment and larger droplets.

In the *inertia-dominated* regime, which applies to low-viscosity systems such as water-in-oil emulsions, droplet detachment occurs more rapidly, and t_p follows:

$$t_p \sim \sqrt{\frac{\rho_d \, r_0^3}{\sigma}},$$

where ρ_d denotes the density of the dispersed phase. Here, capillary instabilities drive the breakup rather than viscous damping.

Experimentally, the pinch-off time is often observed to scale inversely with the local pressure differential and the mean flow velocity:

$$t_p \propto \frac{1}{\Delta P} \quad \text{or} \quad t_p \propto \frac{1}{U_c}.$$

Therefore, increasing actuation voltage or operating frequency—which both elevate ΔP and U_c—reduces t_p and promotes faster droplet formation.

Although precise fluid properties were not measured experimentally in this work, the above relationships provide a consistent physical framework for understanding the droplet formation behavior under varying flow and actuation conditions. The generated pressure differential ΔP from the micropump and the overall hydraulic resistance R_h of the tubing and microchannel jointly determine the resulting droplet volume V_d, which can be expressed as:

$$V_d \sim \frac{\Delta P}{R_h} \, t_p.$$

Hence, higher driving voltages (increasing ΔP) or reduced flow resistance directly contribute to smaller droplets and higher generation rates under otherwise identical conditions.

Results

This chapter presents the experimental characterisation of the developed micropump system and its integration into the droplet generator. The results are organised to reflect the functional chain of the device: from electrical power requirements, to hydraulic performance, and finally to droplet formation. The power-consumption measurements establish the expected battery runtime and indicate whether the system is suitable for portable or incubator-based operation. The subsequent pressure and flow experiments quantify how the electrical actuation translates into usable hydraulic output, defining the operational window in which stable pumping is achievable. Based on this characterisation, the final section examines the behaviour of the two-inlet droplet generator and evaluates the conditions under which stable aqueous and gel droplets can be produced.

4.1 Power Consumption Measurement

A nominal battery capacity of 10 Ah was assumed, leading to an ideal runtime $t = \frac{C}{I}$, where C is the battery capacity and I the total current draw. In practice, a conservative usable capacity of 80% was applied to account for efficiency losses, voltage drop, and battery protection circuitry.

Scenario	**Current (mA)**	**Ideal (h)**	**80% (h)**
No pump active (idle)	200	50.0	40.0
One pump (120 V_{pp}, 200 Hz)	220	45.5	36.4
Three pumps (120 V_{pp}, 200 Hz)	265	37.7	30.2
Three pumps (150 V_{pp}, 500 Hz)	350	28.6	22.9

Table 4.1: Measured current draw and estimated battery runtime under different configurations.

Table 4.1 tells us under ideal conditions, with three pumps operating at 120 V_{pp} and 200 Hz, the system can theoretically operate for approximately **37.7 hours**, or about **30 hours** when accounting for practical losses.

It is important to note that the majority of the current consumption originates not from the micropumps themselves, but from the ESP32 microcontroller and its integrated WiFi transceiver, which exhibits a relatively high idle and active power demand. The actual energy used for driving the piezoelectric elements is comparatively small, as the MP6 drivers operate with low current at high voltage. Therefore, optimizing wireless communication duty cycles and reducing WiFi activity could substantially increase the total operating time in battery-powered mode.

4.2 Pressure Measurement

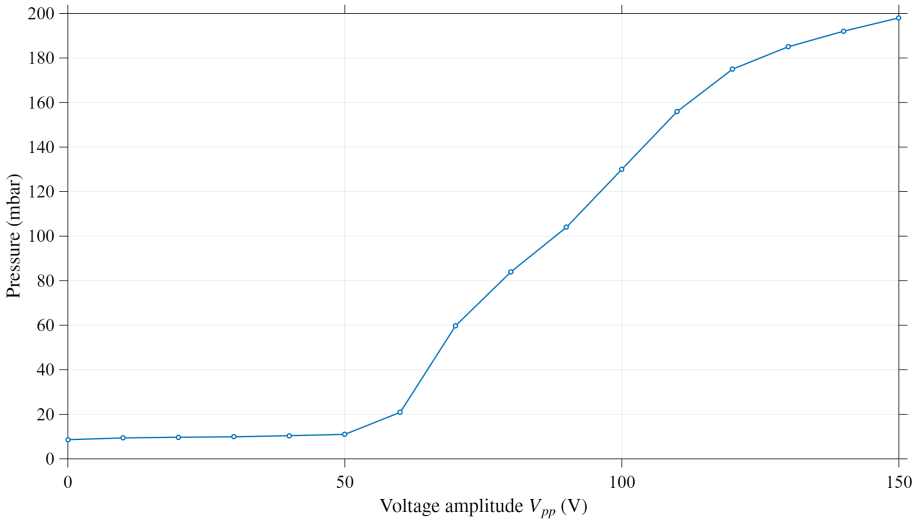


Figure 4.1: Pressure vs. voltage at 200 Hz. No visible Change in Pressure below 50 V

The primary goal of the pressure measurements was to characterize the static (mean) and transient pressure behavior of the Bartels MP6 micropump under various driving voltages and frequencies. The pressure was recorded using a Fluigent Pressure M sensor connected through a 110 mm PTFE tube, while the actuation signals were generated by the custom driver circuitry.

A voltage–pressure characteristic was first obtained at 200 Hz. Figure 4.1 shows a clear threshold-like behavior: below approximately 50 V_{pp}, the measured pressure remains close to the noise floor, while above this value, a nearly linear increase in pressure is observed. Physically, this indicates that the diaphragm motion below 50 V_{pp} is insufficient to overcome the total hydraulic resistance of the setup (tubing, connectors, sensor inlet, and internal check valves). Once the deformation amplitude exceeds this threshold, the generated volume displacement becomes large enough to produce a measurable and steady output pressure.

The behavior in Figure 4.2 corresponds to a low-pass filtering effect of the hydraulic system: the mechanical compliance of the pump chamber, tubing inertia, and the limited bandwidth of the pressure sensor collectively attenuate higher-frequency components. Consequently, at lower frequencies, the transient peaks remain visible, while at higher frequencies they are averaged out. From an application perspective, such transient peaks may be advantageous for pumping higher-viscosity fluids, as the short but intense pressure pulses can temporarily overcome viscous resistance more effectively than a constant low pressure. It should be noted that these pressure peaks are short-lived and do not necessarily translate into sustained flow. For applications requiring continuous pumping or maintenance of stable pressure only the mean pressure is relevant. Moreover, the actual peak values may be underestimated due to the finite response time of the pressure sensor and the sampling rate of the data acquisition system. A higher-bandwidth differential pressure sensor could capture these short transients more accurately.

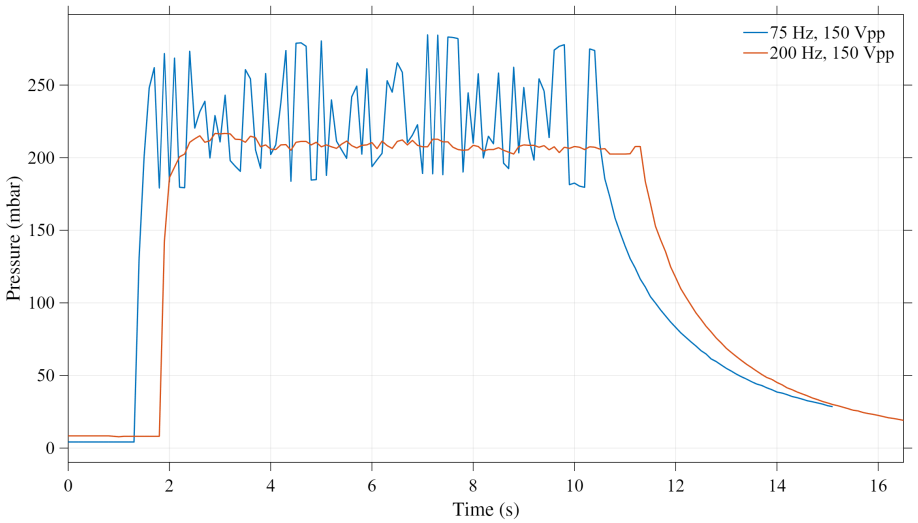


Figure 4.2: Pressure at 75 Hz and 200 Hz. Time-dependent measurements at 75 Hz and 200 Hz provided further insight into the dynamic response. At 200 Hz, the pressure signal appears smoother and more averaged, as the rapid actuation cycles are integrated by the hydraulic system and sensor response, resulting in a relatively stable mean pressure. In contrast, at 75 Hz, distinct pressure spikes are visible, where the instantaneous pressure briefly exceeds the mean value by a significant margin.

4.3 Flow Measurement

Flow characterization was performed under the same electrical driving conditions as the pressure measurements, using a **Fluigent Flow Unit M** with a maximum measurable range of 80 μL/min. The flow rate was recorded over time at different peak-to-peak voltages (V_{pp}) and driving frequencies (f), providing both instantaneous and averaged flow profiles for each operating condition.

Figure 4.3 shows the instantaneous flow traces for all voltage–frequency combinations. At low driving frequencies such as 25 Hz, the flow signal exhibits clear pulsatile peaks that directly reflect the membrane actuation cycles. These peaks reach the 80 μL/min sensor limit at relatively low voltages, which makes accurate measurements above approximately 50 $\mathrm{V_{pp}}$ infeasible in this regime. At higher frequencies, the pulsatile components are increasingly attenuated, leading to smoother time-domain flow profiles consistent with the low-pass hydraulic behavior observed in the pressure measurements.

The averaged flow rates are summarized in Figure 4.4. Across all frequencies, the mean flow increases approximately linearly with applied voltage until the sensor saturation limit is reached. The highest mean flow values were consistently obtained at 150 Hz and 200 Hz, while lower frequencies (25–50 Hz) produced substantially weaker flow. This trend reflects the increasing smoothing of the pulsatile flow at higher frequencies, leading to a more continuous and effective net displacement per unit time.

The lowest stable mean flow was approximately 3–5 μL/min across all tested conditions. Notably, this minimum did not occur at the lowest frequency but rather emerged from the interplay between actuation frequency, voltage, and the hydraulic filtering properties of the system. Since several high-voltage measurements at mid and high frequencies approach or exceed the 80 μL/min limit, the true maximum flow rate of the pump is likely higher than the measurable range of the available flow sensor.

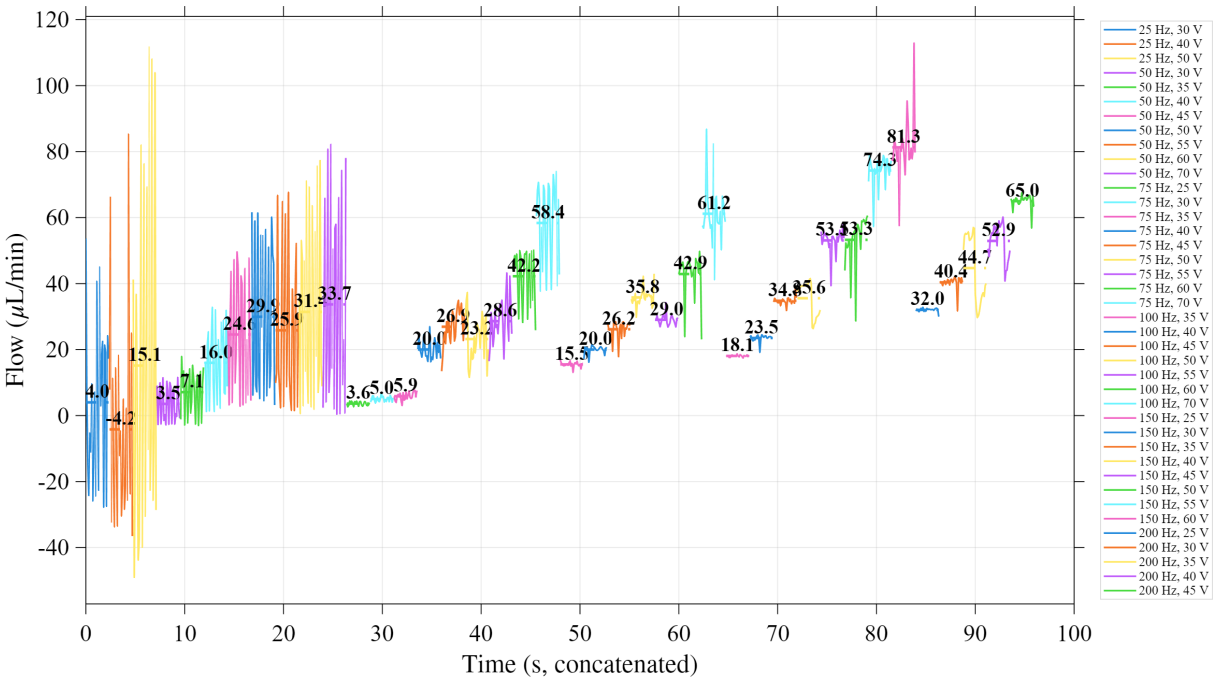


Figure 4.3: Complete time-resolved flow data for all tested operating conditions. At low frequencies (e.g., 25 Hz), distinct oscillatory peaks are visible, corresponding to the periodic actuation of the piezoelectric diaphragm.

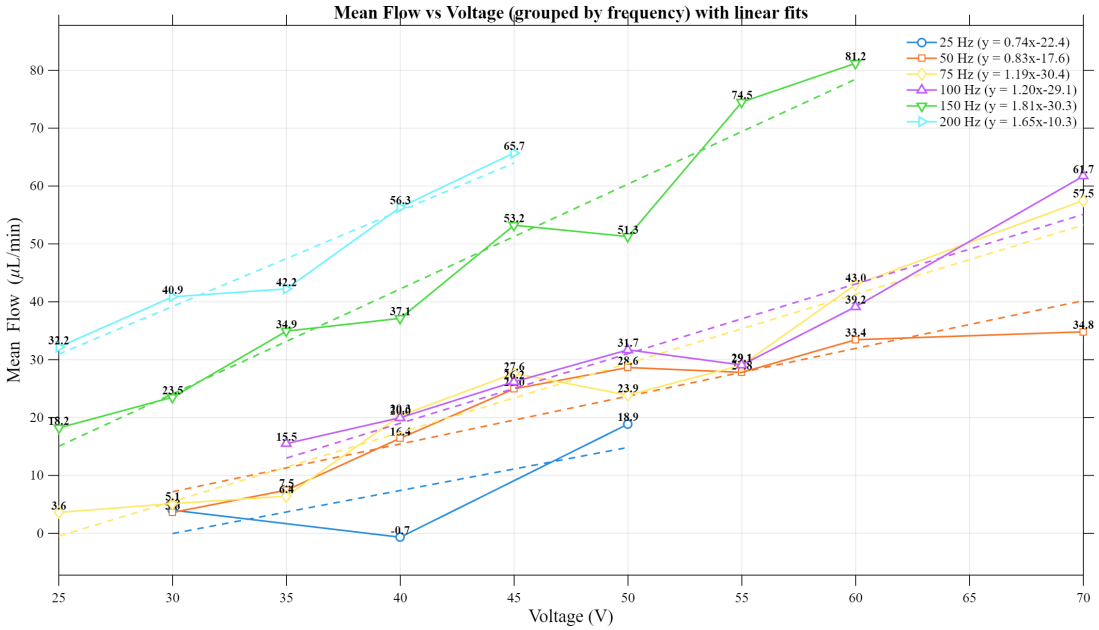
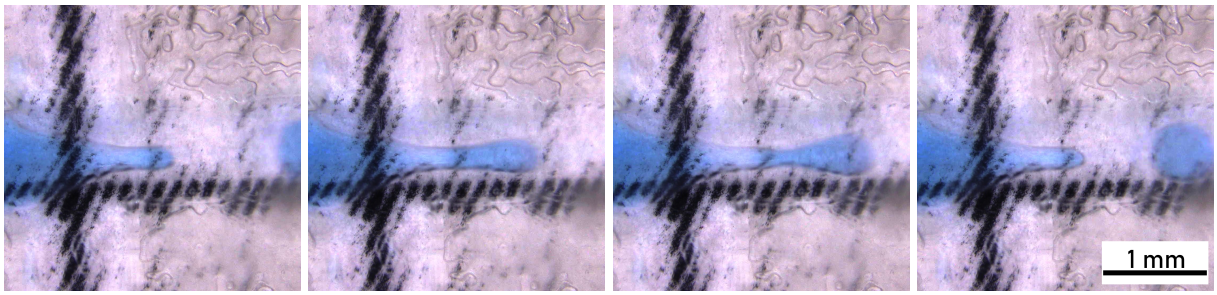


Figure 4.4: Mean flow rate as a function of voltage, grouped by frequency, including the corresponding linear fits.

4.4 Droplet Generation

Having the micropump characterized, microfluidic droplets were generated next with distilled water and CMC gel solutions of varying concentrations as the dispersed phase, with rapeseed oil as the continuous phase.

Due to the hydraulic resistance of the microchannel system, fluids with higher viscosity,



(a) Initial state (b) Nucleation (c) Separation (d) Stable droplet

Figure 4.5: Microscopic images showing the formation process of a droplet (a–d).

such as oils or CMC gels, could only be effectively transported at lower driving frequencies and higher voltage amplitudes, exploiting the transient pressure peaks generated by the piezoelectric actuation. With the present setup, successful pumping and droplet formation were achieved with CMC concentrations up to 1 wt%, corresponding to a viscosity of approximately $530\ mPa \cdot s$ at zero shear rate.

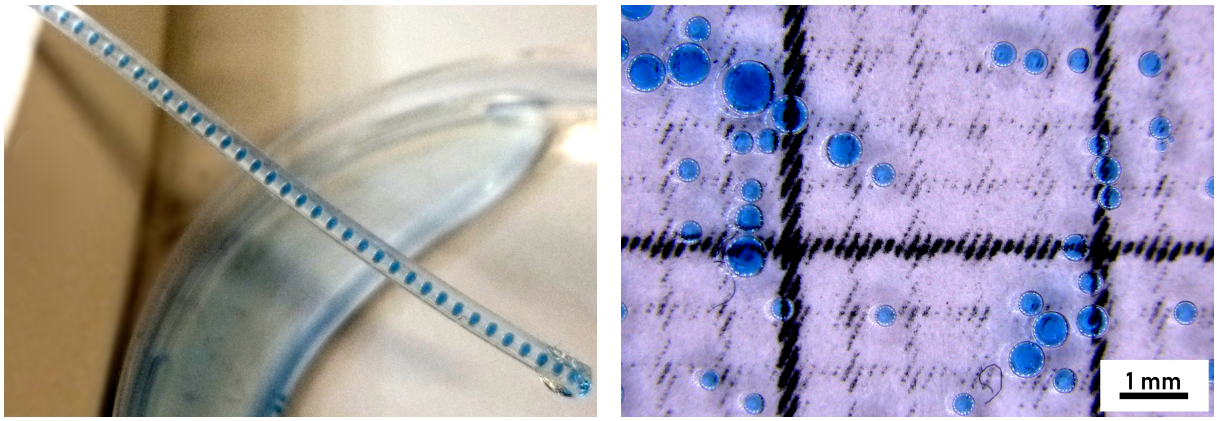
For water, the droplet generation was consistent and highly stable across all parameter ranges where droplet detachment could be observed. The droplet size and detachment frequency exhibited an approximately linear dependence on the pressure difference between the two phases, confirming the controllability of the system within the laminar regime.

In contrast, the CMC gel solutions displayed a more complex behavior due to their shear-thinning rheological properties . Despite this, stable droplet formation was achieved for concentrations of 0.2, 0.5, and 1 wt %. However, with increasing concentration, the solution nonlinearity became more pronounced, resulting in droplet sizes that were strongly dependent on local flow conditions and transient shear states. As shown in Figure 5.1, stable and periodic droplet formation was achieved for the 0.2 wt % gel solution (oil phase: 120 V_{pp}, 40 Hz; gel phase: 80 V_{pp}, 70 Hz). For more concentrated gels, even minor air inclusions or particulate contaminants led to pronounced instabilities, oscillations, or elongated droplet shapes. In some cases, the droplet pinch-off location shifted noticeably within the microchannel, indicating sensitivity to flow boundary variations.

Finally, the generated droplets remained stable after detachment, as observed in the Petri dish collection tests (see Figure 5.1), confirming the mechanical integrity of gel droplets within the oil medium.

Discussion and Conclusion

This section evaluates the behaviour of the developed system, examines the key factors influencing its performance and outlines the most relevant directions for further improvement.



(a) Stabilty (0.2 wt% CMC) (b) Collected droplets in Petri dish

Figure 5.1: Visualization of droplet formation and stability under different conditions.

The development of the piezoelectric micropump and the 3D-printed droplet generator was guided by the goal of creating a versatile, modular platform for low-flow microfluidic applications. Several design aspects were optimized to ensure general applicability and compatibility with a variety of liquid media; however, the present thesis was limited to demonstrating the system's feasibility rather than a full performance optimization. Figure 5.1 illustrates two key aspects of the system's behaviour. The left panel shows stable and periodic droplet formation for a 0.2 wt% CMC solution under suitable operating conditions, confirming that the dual-inlet configuration can sustain regular pinch-off. The right panel displays droplets collected in a Petri dish. Although the droplets do not all exhibit identical size or shape, they remain mechanically stable after detachment.

5.1 System Behavior and Practical Considerations

During experimentation, it became evident that each micropump exhibited an individual flow and pressure characteristic. This variation can be attributed to small differences in fabrication tolerances, residual contamination, or air entrapment in the internal channels. Therefore, prior to any measurement or application, each pump should be treated as an

individual unit, calibrated, and pre-conditioned. A practical pre-operation routine was established to ensure consistent performance: before filling the system with the working fluid, each pump should be operated for several seconds at approximately 500 Hz and 150 V_{pp} to release possible contaminants or microbubbles adhered to the membrane. Subsequently, the pump and tubing should be flushed with distilled water before introducing more viscous fluids such as CMC gels. This procedure was found to significantly improve flow stability and reproducibility and reduce the overall process time.

The current parameter space covered only a subset of the pump's operational range. For a more complete characterization, both the pressure and flow behavior should be mapped across the entire control domain of actuation voltage and frequency. Such data would provide valuable insight into the dynamic response and potential nonlinearities of the system. The operational range at higher pressure and flow might be increased by using multiple pumps in series or parallel, although this was not evaluated here. The lower limit of stable flow, observed between 3 and 5 µL/min, is compatible with most biological or analytical microfluidic applications. To assess long-term performance, continuous operation and monitoring over a 24-hour period would be advisable, enabling the evaluation of drift, fatigue, and flow stability.

5.2 Summary of Findings

The characterisation established several quantitative limits of the system. The lowest stable flow rates were in the range of 3–5 µL/min and did not occur above 100Hz driving frequency. The flow sensor saturated at 80 µL/min, indicating that the pump output exceeded the measurable range. Under realistic operating conditions, the system achieved a practical battery runtime of approximately 30 hours.

Operation of the three-inlet configuration highlighted the main limitation of the current microfluidic setup. Differences in hydraulic resistance, residual air and small variations in pump behaviour produced asynchronous pressure responses, which disrupted the balance between the inlet streams. As a result, the two continuous-phase pumps occasionally counteracted each other rather than maintaining coordinated flow, preventing stable droplet formation in the three-inlet geometry.

The results demonstrate that the combination of a piezoelectric micropump system and a 3D-printed dual-inlet droplet generator is capable of reliably producing microdroplets. The system permits adjustment of key formation parameters such as actuation frequency, voltage and fluid viscosity. Stable droplet generation was obtained for water and for CMC solutions up to 1 wt% in an oil continuous phase, confirming the basic functionality of the platform, although without reaching a fully optimised operating regime.

5.3 Outlook

Future work should prioritise improving the synchronisation of multiple pumps through closed-loop control. Incorporating real-time flow or pressure feedback would allow adaptive compensation for pump-to-pump variability and dynamic load changes, thereby stabilising multi-inlet operation.

Electrical measurements such as pump current and output voltage may serve as an additional diagnostic tool, offering optional insight into individual pump behaviour and supporting fine-tuning of operating conditions when required.

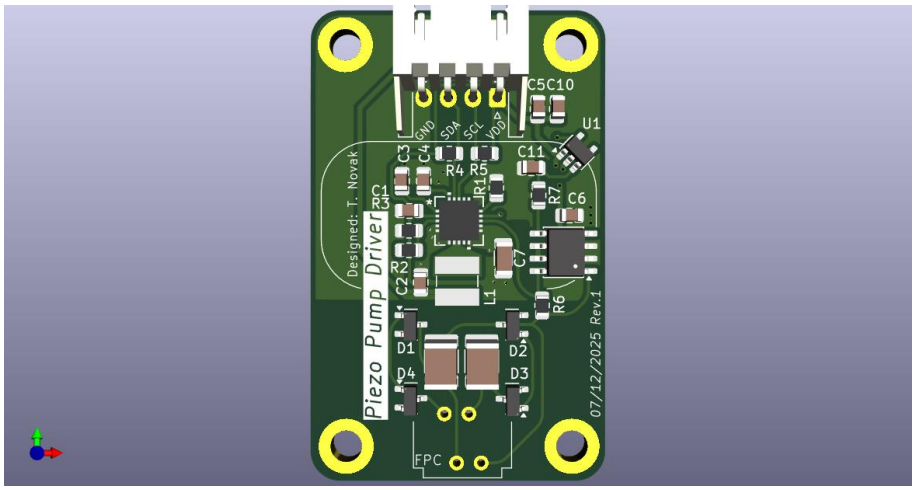


Figure 5.2: Stand-alone piezo pump driver - KiCAD 3D View

Finally, developing a fully self-contained pump-driver module capable of generating high-voltage waveforms, acquiring sensor data, and executing control algorithms autonomously would greatly enhance usability and enable deployment in portable or application-specific microfluidic platforms.

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Appendix

The completed codes, circuit designs, and 3D files can be found in the following GitHub repository: https://github.com/Schimpfen/CCG_UPump/tree/main

A.1 System Overview

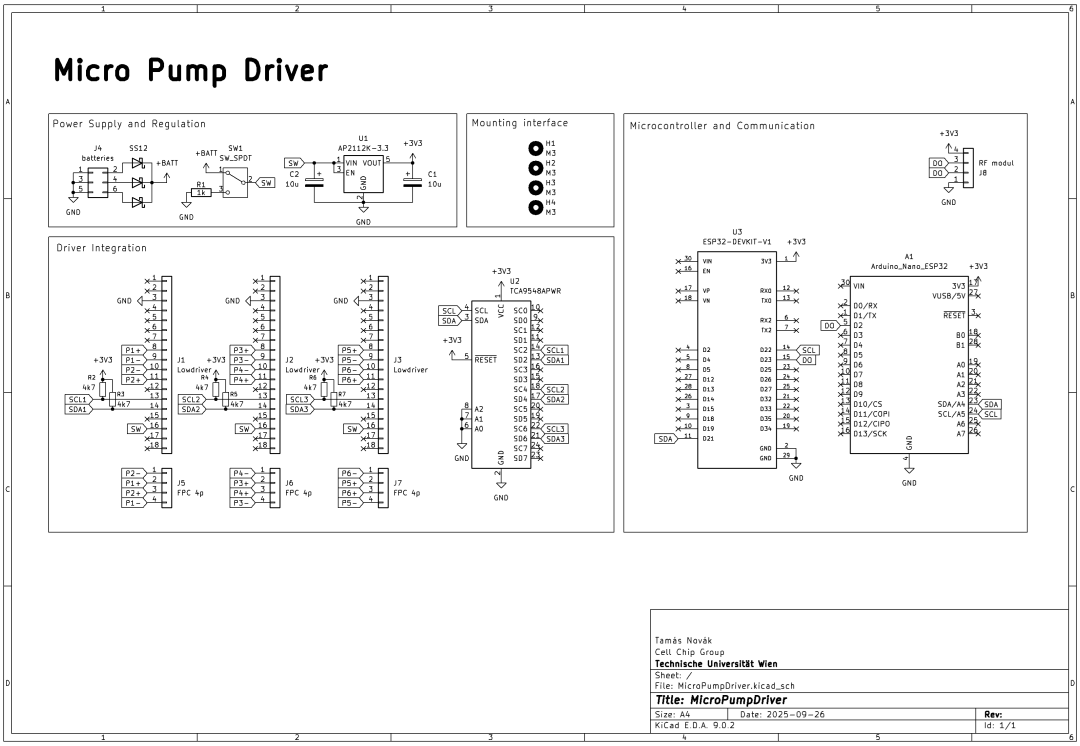


Figure A.1: Schematic view

A.2 Validity of Flow Measurement

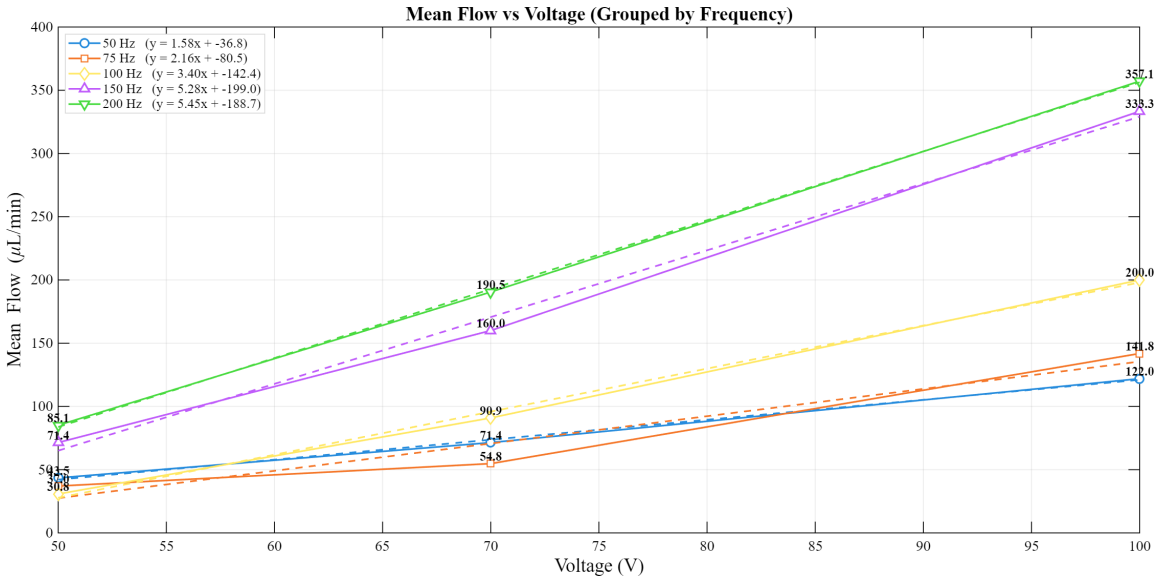


Figure A.2: The diagram below shows the results of a validation flow measurement. It presents the amounts of distilled water of known volume pumped per unit time at various voltage and frequency settings.

A.3 Mainboard for stand-alone pumps

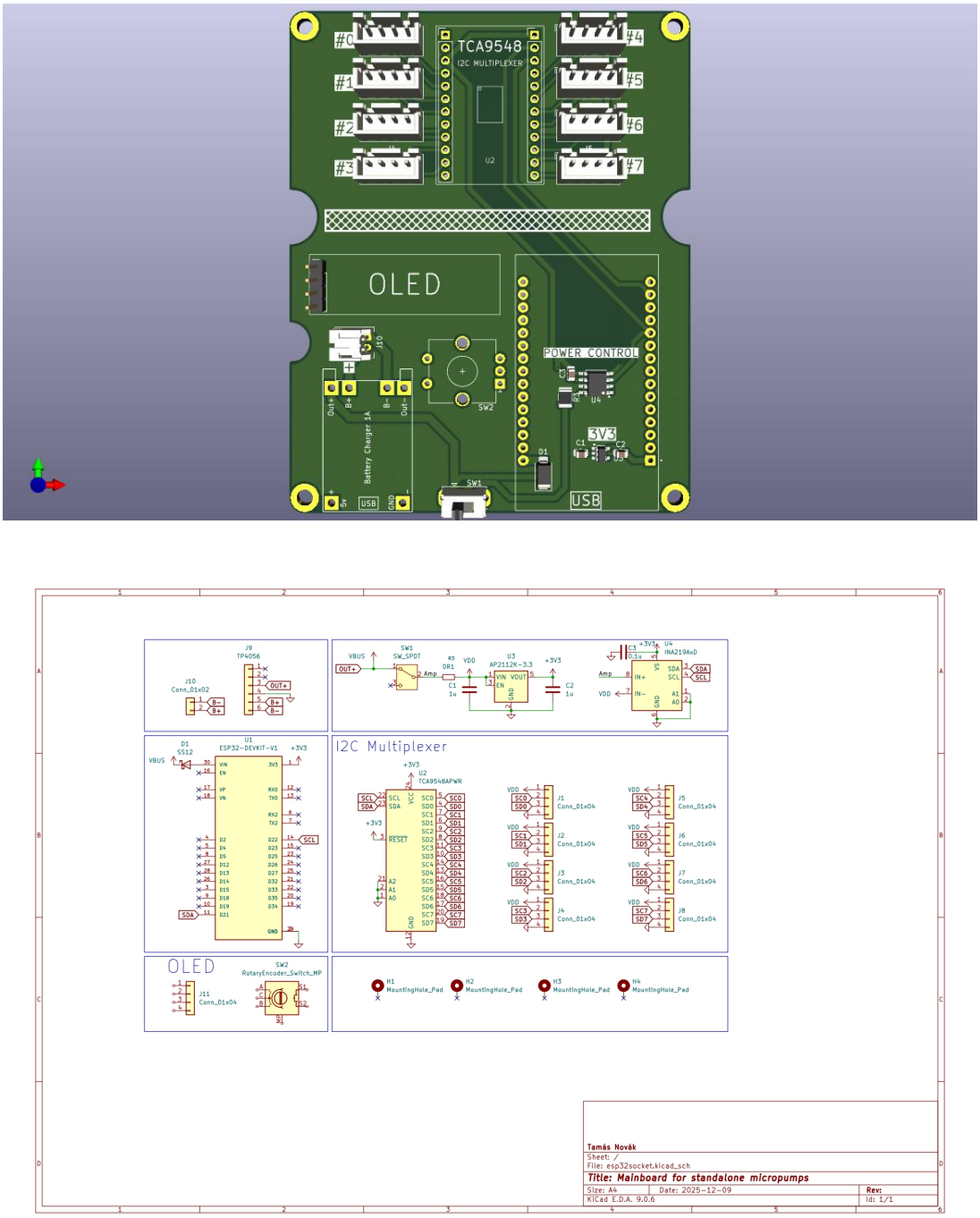


Figure A.3: The central board under development is capable of driving eight individual pumps simultaneously and includes a battery voltage and power monitor that can trigger warning actions.

A.4 Driver schematic

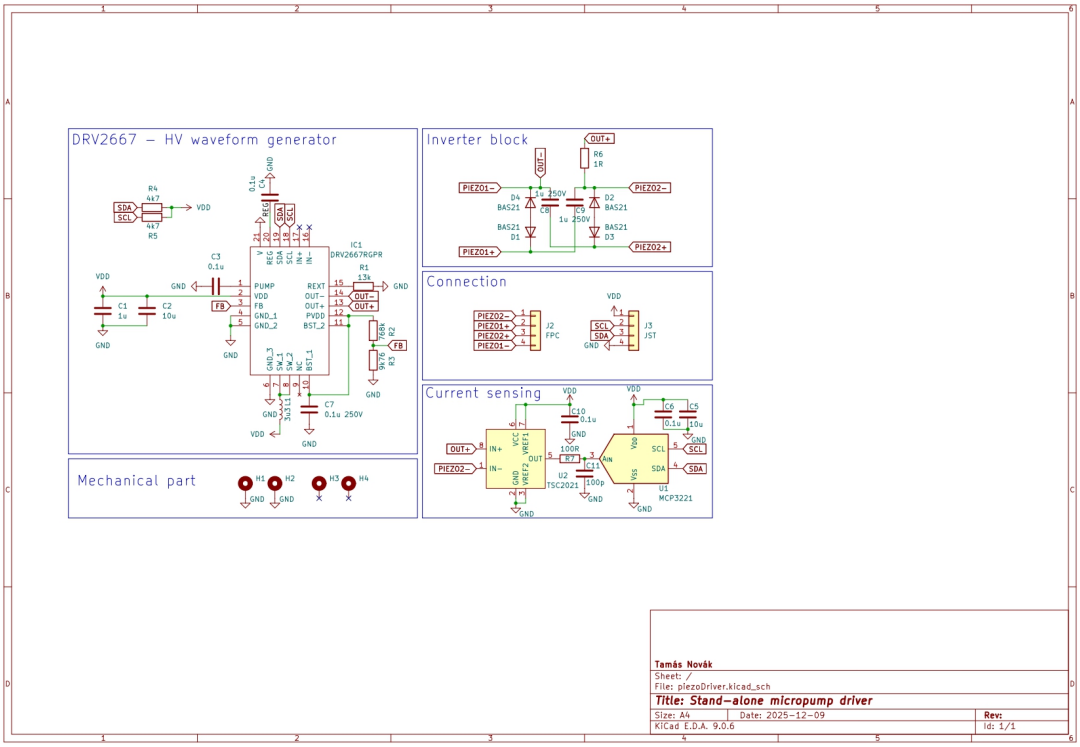


Figure A.4: A custom piezo pump driver circuit layout that incorporates a high-voltage waveform generator and current-curve feedback.